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Devices, systems, and methods for clamping a leaflet of a heart valve

Abstract

The present disclosure relates generally to the field of medical devices for clamping a leaflet of a heart valve. In particular, the present disclosure relates to medical devices, systems, and methods for delivering artificial chordae tendineae in a patient. In an embodiment, a system may include a clamp having a plurality of arms at a first end. The plurality of arms may have a closed configuration in which the arms are oriented toward each other, and an open configuration in which the arms are oriented away from each other. A spring portion may be coupled to the plurality of arms at a second end that is configured to bias the arms to the closed configuration. The arms of the clamp may be configured to fixedly engage with a leaflet of the heart valve. The second end of the clamp may be configured to couple to an artificial chordae tendineae.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. application Ser. No. 16/919,782, filed Jul. 2, 2020, which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application 62/873,354, filed Jul. 12, 2019, both of which applications are incorporated herein by reference in their entirety for all purposes.

FIELD

(1) The present disclosure relates generally to the field of medical devices for clamping a leaflet of a heart valve. In particular, the present disclosure relates to medical devices, systems and methods for delivering artificial chordae tendineae in a patient.

BACKGROUND

(2) Mitral valve disease is typically repaired via invasive surgical intervention or by complicated pinching of the leaflets together creating dual, smaller openings, or a mitral valve replacement of the native valve. These approaches involve risky by-pass surgery that may include an opening into the patient's chest and heart chamber to expose the mitral valve for direct viewing and repair. Resection, partial removal, and/or repair of the patient's leaflets along with the implantation of a surgical ring are complex techniques used by surgeons to reduce the diameter of the patient's mitral annulus, thus allowing the leaflets to properly coapt and reduce mitral regurgitate flow. Some techniques may slightly reduce regurgitate flow but may not provide a durable solution and do not repair and/or replace damaged chordae tendineae of a valve. Thus, transluminal solutions to mitral valve disease are needed.

(3) A variety of advantageous medical outcomes may be realized by the medical devices, systems, and methods of the present disclosure, which include clamping of a leaflet of a heart valve.

SUMMARY

(4) Embodiments of the present disclosure may assist generally with clamping of a heart valve and providing a connection point for a filament. In one aspect, a system for clamping a leaflet of a heart valve may include a clamp. The clamp may include a plurality of arms at a first end. The plurality of arms may have a closed configuration in which the arms are oriented toward each other, and an open configuration in which the arms are oriented away from each other at an open distance between the arms that is greater than a closed distance between the arms in the closed configuration. The closed distance may be zero millimeters. A spring portion may be coupled to the plurality of arms at a second end. The spring portion may be configured to bias the arms to the closed configuration. The arms of the clamp may be configured to fixedly engage with a leaflet of the heart valve. The second end of the clamp may be configured to couple to an artificial chordae tendineae.

(5) In various embodiments described here or otherwise, a spreader may be configured to transition the clamp between the closed configuration and the open configuration. The spreader may include a base. A pin may extend from the base. A lever may be rotatably disposed about the pin. A first channel may extend through the base substantially parallel with a first aperture of one of the plurality of arms and may be configured to accept one of the plurality of arms. A second channel may extend through the lever substantially parallel with a second aperture of one of the plurality of arms and may be configured to accept one of the plurality of arms. A first filament may extend from the lever. The filament may be configured to move the lever and the clamp between the closed

configuration and the open configuration. A catheter may have a distal end of the catheter coupled to the base. A first pin may be disposed within the first aperture and the first channel. A second pin may be disposed within the second aperture and the second channel. A second filament may couple the first pin to the second pin. One or more protrusions may be disposed on one or more of the plurality of arms. The one or more protrusions may be selected from the group consisting of barbs, spikes, hooks, and tines. The one or more protrusions may be a plurality of protrusions arranged in columns extending along at least one of the plurality of arms such that the plurality of protrusions are in different planes. The protrusions may extend not more than 50% through a thickness of a wall of the leaflet, e.g., the protrusions may not extend into the wall of a leaflet and may instead distort the tissue. The protrusions may extend a distance from an arm that may be about 0.5 millimeters to about 1.5 millimeters. The clamp may weigh less than 0.08 grams.

(6) In an aspect, a clamp for clamping a leaflet of a heart valve may include a body. The body may include a plurality of arms at a first end. The plurality of arms may have a closed configuration in which the arms are oriented toward each other, and an open configuration in which the arms are oriented away from each other. The arms of the clamp may be configured to fixedly engage with a leaflet of the heart valve. A second end of the clamp may be configured to couple to an artificial chordae tendineae.

(7) In various embodiments described here or otherwise, the body may include a coiled spring portion at a second end. The spring portion may be configured to bias the arms to the closed configuration. A longitudinal axis may extend through the first end and the second end. The clamp may include a first cover disposed adjacent to a first arm of the plurality of arms and extending along the first arm to the longitudinal axis. A second cover may be disposed adjacent to a second arm of the plurality of arms and extending along the second arm to the longitudinal axis. A pin may extend through the first cover, the second cover, and the coiled spring portion. A first channel may be disposed on the first cover having a central axis that is substantially parallel with the first arm. A second channel may be disposed on the second cover having a central axis that is substantially parallel with the second arm. The first and second channels may be configured to substantially align with second and third apertures of a spreader. A first pin may extend into the first channel and the second aperture. A second pin may extend through the second channel and the third aperture. A filament coupling the first pin to the second pin. A tab may be configured to transition the arms between the open configuration and the closed configuration. The open configuration and the closed configuration are may each be stable configurations.

(8) In an aspect, a method of clamping a leaflet of a heart valve may include inserting a catheter through the valve. The catheter may include a spreader disposed on a distal end of the catheter and reversibly coupled to a clamp. The clamp and spreader may be positioned proximate to the leaflet. Tension on a first filament extending through the catheter may be released and connected to the spreader such that the clamp transitions to a closed configuration about the leaflet. A plurality of pins may be removed from the spreader and clamp such that the spreader releases the clamp. Tension may be applied to the first filament such that the spreader transitions the clamp from the closed configuration to an open configuration. The clamp may be repositioned about the leaflet. Tension may be released on the first filament extending through the catheter such that the clamp transitions to the closed configuration about the leaflet. An artificial chordae tendineae attached to the clamp may be anchored to a papillary muscle. The leaflet may be a flailing leaflet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting embodiments of the present disclosure are described by way of example with reference to the accompanying figures, which are schematic and not intended to be drawn to scale.

In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment shown where illustration is not necessary to allow those of ordinary skill in the art to understand the disclosure. In the figures:

- (2) FIG. 1 illustrates a cross-sectional view of flailing leaflet of a mitral valve during blood flow regurgitation.
- (3) FIG. 2A illustrates a perspective view of a clamp in a closed configuration, according to an embodiment of the present disclosure.
- (4) FIG. 2B illustrates another perspective view of the clamp in FIG. 2A.
- (5) FIG. 2C illustrates a side view of the clamp in FIGS. 2A and 2B.
- (6) FIG. 2D illustrates a perspective view of the clamp in FIGS. 2A-2C in an open configuration.
- (7) FIG. 2E illustrates a side view of the clamp in FIGS. 2A-2D.
- (8) FIG. 3A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (9) FIG. 3B illustrates a side view of the clamp in FIG. 3A.
- (10) FIG. 4A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (11) FIG. 4B illustrates a side view of the clamp in FIG. 4A.
- (12) FIG. 5A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (13) FIG. 5B illustrates a side view of the clamp in FIG. 5A.
- (14) FIG. 6A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (15) FIG. 6B illustrates a side view of the clamp in FIG. 6A.
- (16) FIG. 7A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (17) FIG. 7B illustrates a front view of the clamp in FIG. 7A.
- (18) FIG. 8A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (19) FIG. 8B illustrates a front view of the clamp in FIG. 8A.
- (20) FIG. 9A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (21) FIG. 9B illustrates a front view of the clamp in FIG. 9A.
- (22) FIG. 10A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (23) FIG. 10B illustrates a side view of the clamp in FIG. 10A.
- (24) FIG. 11A illustrates a perspective view of a clamp, according to an embodiment of the present disclosure.
- (25) FIG. 11B illustrates a top view of the clamp of FIG. 11A.
- (26) FIG. 11C illustrates a bottom view of the clamp of FIGS. 11A and 11B.
- (27) FIG. 12A illustrates a side view of a clamp in a closed configuration, according to an embodiment of the present disclosure.
- (28) FIG. 12B illustrates the clamp of FIG. 12A in an open configuration.
- (29) FIG. 13A illustrates a perspective view of a system for clamping a leaflet of a heart valve in a closed configuration, according to an embodiment of the present disclosure.
- (30) FIG. 13B illustrates a side view of the system of FIG. 13A.
- (31) FIG. 13C illustrates the system of FIGS. 13A and 13B disposed on a catheter in an open configuration.
- (32) FIG. 14A illustrates a cross-sectional view of a clamp clamping a leaflet of a heart valve, according to an embodiment of the present disclosure.

(33) FIG. 14B illustrates another cross-sectional view of the clamp and leaflet of FIG. 14A.

(34) FIG. 14C illustrates another cross-sectional view of an alternative orientation of the clamp and leaflet of FIG. 14A.

(35) FIG. 15 illustrates a cross-sectional view of a system for clamping a leaflet, according to an embodiment of the present disclosure.

(36) FIG. 16 illustrates a perspective view of the clamp in FIGS. 2A-2E having alternative protrusions, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(37) The present disclosure is not limited to the particular embodiments described. The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting beyond the scope of the appended claims. Unless otherwise defined, all technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs.

(38) Although embodiments of the present disclosure may be described with specific reference to medical devices and systems (e.g., transluminal devices inserted through a femoral vein or the like) for selective access to a heart valve, it should be appreciated that such medical devices and systems may be used in a variety of medical procedures that require clamping a leaflet of a valve or clamping a tissue wall. The disclosed medical devices and systems may also be inserted via different access points and approaches, e.g., percutaneously, endoscopically, laparoscopically, or combinations thereof.

(39) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises” and/or “comprising,” or “includes” and/or “including” when used herein, specify the presence of stated features, regions, steps, elements and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components and/or groups thereof.

(40) As used herein, “proximal end” refers to the end of a device that lies closest to the medical professional along the device when introducing the device into a patient, and “distal end” refers to the end of a device or object that lies furthest from the medical professional along the device during implantation, positioning, or delivery.

(41) As used herein, the conjunction “and” includes each of the structures, components, features, or the like, which are so conjoined, unless the context clearly indicates otherwise, and the conjunction “or” includes one or the others of the structures, components, features, or the like, which are so conjoined, singly and in any combination and number, unless the context clearly indicates otherwise.

(42) All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about”, in the context of numeric values, generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure. Other uses of the term “about” (i.e., in a context other than numeric values) may be assumed to have their ordinary and customary definition(s), as understood from and consistent with the context of the specification, unless otherwise specified. The recitation of numerical ranges by endpoints includes all numbers within that range, including the endpoints (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

(43) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments, whether or

not explicitly described, unless clearly stated to the contrary. That is, the various individual elements described below, even if not explicitly shown in a particular combination, are nevertheless contemplated as being combinable or arrangeable with each other to form other additional embodiments or to complement and/or enrich the described embodiment(s), as would be understood by one of ordinary skill in the art.

(44) Heart disease including atrioventricular heart valve malfunctions impede patient cardiac output, which reduces patient quality of life and lifespan. With reference to the heart **154** illustrated in FIG. **1**, as heart disease progresses, the chordae tendineae **155** that connect the papillary muscle **152** to a valve leaflet **151** may stretch inelastically and may rupture. A stretched and/or ruptured chordae tendineae **156** may result in a flailing leaflet **150** that may no longer have capacity to form a valving seal for normal heart function. For example, abnormal blood flow regurgitation in the direction of vector **158** may develop. Regurgitation prevents an adequate supply of blood to be delivered through the cardiovascular systems.

(45) Repositioning, repair, and/or replacement of one or more leaflets of a valve and/or chordae tendinea may be desirable to treat heart disease. The devices, systems, and methods of the present disclosure may be used alone or together with other devices, systems, and methods to treat heart disease. Examples of devices, systems, and methods with which embodiments of the present disclosure may be implemented include, but are not limited to, those described in U.S. patent application Ser. No. 16/919,769, filed Jul. 2, 2020, and titled Devices, Systems, and Methods for Adjustably Tensioning an Artificial Chordae Tendineae Between a Leaflet and a Papillary Muscle or Heart Wall; U.S. patent application Ser. No. 16/919,806, filed Jul. 2, 2020, and titled Devices, Systems, and Methods for Artificial Chordae Tendineae; and U.S. patent application Ser. No. 16/919,794, filed Jul. 2, 2020, and titled Devices, Systems, and Methods for Anchoring an Artificial Chordae Tendineae to a Papillary Muscle or Heart Wall, each of which is filed on even date herewith and each of which is herein incorporated by reference in its entirety and for all purposes. Examples of devices described therein may be modified to incorporate embodiments or one or more features of the present disclosure. Repositioning, repair, and/or replacement of one or more leaflets of a valve and/or chordae tendinea may include one or more devices to be fixed to one or more leaflets of a valve. Embodiments of devices described herein may be fixed to a valve by clamping to a leaflet. These devices may provide a fixed point for other devices, systems, or tools to grab or attach to in order to manipulate a leaflet of a valve and/or deliver devices attached to the leaflet.

(46) With reference to FIGS. **2A-2E**, an embodiment of a clamp **200** for clamping a leaflet of a heart valve according to the present disclosure is illustrated, which includes two arms **202** at a first end **200p** of the clamp **200**. The arms **202** are in the closed configuration in FIGS. **2A-2C** in which the arms **202** are oriented toward each other. A spring portion **204** may be at a second end **200d** of the clamp **200**. The spring portion **204** may be configured to bias the arms **202** to the closed configuration. The spring portion **204** at the second end **200d** is wider than a width of the clamp **200** at the location of the arms **202** to reduce stress on the clamp **200** while transitioning between the closed configuration and an open configuration. Each of the arms **202** has an aperture **208** at an end of each arm **202**. Each aperture **208** has a central axis **208a** that extends substantially along each of the arms **202** and is configured to accept a locking pin for manipulating the clamp **200** between the closed configuration and the open configuration (as will be discussed below with respect to the discussion of FIGS. **13A-15**). Each arm **202** includes protrusions **206** that extend along each arm **202** that are configured to embed into tissue. Each of the protrusions has a smooth surface **206p** toward the first end **200p** of the clamp **200** that is configured to accept a tissue moving between the arms **202** from the first end **200p** toward the second end **200d**. Each protrusion **206** also includes an engaging end **206d** toward the second end **200d** that is configured to embed into a tissue between the arms **202**. Because the engaging end **206d** of each protrusion **206** may be angled toward the second end **200d** of the clamp **200**, the protrusions **206** may resist a movement

of a tissue between the arms **202** with respect to the clamp **200** by embedding the engaging ends **206d** into the tissue and/or providing a frictional force to the tissue. FIGS. 2D and 2E illustrate the clamp **200** in the open configuration, in which the arms **202** are oriented away from each other. The open configuration can more easily accept tissue (e.g., a leaflet) and can be more easily repositioned to clamp tissue than the closed configuration. The open configuration has the arms **202** apart from each other about an arc length α that may be a various amount of degrees, e.g., about 50°, about 60°, about 90°, etc. The arc length α may be sufficient to enable the clamp **200** to surround at least a portion of a tissue of a valve. A first thickness **210p** of the first end **200p** of the clamp may be substantially equivalent to a second thickness **210d** of the second end **200d**.

Alternatively, the first thickness **210p** may be different than the second thickness **210d** and this difference in thicknesses may gradually change from the first thickness **210p** to the second thickness **210d**. For example, the first thickness **210p** may be thinner than the second thickness **210d** or vice versa. For example, one or more thickness **210p**, **210d** of the clamp may be about 0.1 mm to about 3.0 mm such as a first thickness **210p** of about 0.7 mm and a second thickness **210d** of about 0.1 mm to about 3.0 mm. A width **212** of a clamp may be substantially uniform or may vary along the clamp and may be, e.g., about 2 mm to about 20 mm. A length **214** of a clamp may be, e.g., about 5 mm to about 20 mm. A height **216** of a clamp may be, e.g., about 0.7 mm to about 10 mm.

(47) In various embodiments of the present disclosure, one or more arms of a clamp may have locking features for use with a device that may transition the clamp between a closed configuration and an open configuration. The locking features may include apertures, edges for channels, tabs, or the like. These locking features may be engaged by additional devices such as locking pins, channels, clamps, or the like. The additional devices may engage the locking features to transition the clamp between the closed and open configurations and the additional devices may disengage the locking features to deliver the clamp from one or more devices into a patient.

(48) With reference to FIGS. 3A and 3B, an embodiment of a clamp for clamping a leaflet of a heart valve according to the present disclosure is illustrated, which includes two arms **301**, **302** at a first end **300p** of the clamp. The arms **301**, **302** can transition between a closed configuration in which the arms **301**, **302** are oriented toward each other, and an open configuration in which the arms **302** are oriented away from each other at a distance between the arms **301**, **302** away from each other that is greater than a distance between the arms **301**, **302** in the closed configuration. A spring portion **304** at a second end **300d** of the clamp is configured to bias the arms **301**, **302** to the closed configuration. One arm **301** includes protrusions **306** that extend along the arm **302** that are configured to embed into tissue. Each of the protrusions **306** may be angled toward the second end **300d** of the clamp **300** to resist a movement of a tissue between the arms **301**, **302** with respect to the clamp by embedding the protrusions **306** into the tissue and/or providing a frictional force to the tissue. One of the arms **302** includes a slot **307** extending along a portion of the length of the arm **302** that may assist with the protrusions **306** embedding into tissue by providing a window defined by struts of the arm **302** about the protrusions **306** and/or may lighten the weight of the clamp **300** by reducing the material used compared to a clamp without a vacant slot **307**. In FIGS. 4A and 4B, the protrusions **306** on an arm **301** are angled perpendicularly to the arm **301**. In FIGS. 5A-6B, the protrusions **306** on one arm **302** extend along a longitudinal axis of the arm **302**, while the protrusions **305** on the opposing arm **301** extend in two spaced apart columns adjacent to the longitudinal axis of the opposing arm **301**. The protrusions **305** may be oriented to oppose a direction the clamp is likely to slide (e.g., oriented toward an end of a leaflet). In some embodiments, the protrusions **305** may each be in a different plane from each other to reduce tearing propagation from adjacent protrusions **305**. In some embodiments, the protrusions may be barbs, spikes, hooks, tines, or the like.

(49) In various embodiments, a clamp may be manufactured to be made up of a material adequate to provide a sufficient clamping force to fixate the clamp to a leaflet and provide a substantial

anchoring body for an attached filament such as an artificial chordae tendineae. In an aspect, a clamp may be manufactured to provide sufficient force to enable the clamp to fixedly attach to the valve tissue without imparting undue gravitational forces that may disrupt efforts of valve repair. In this way, the clamp has a mass that is not significantly larger than necessary such that the weight of the clamp does not negatively impact the leaflet or nearby tissue. For example, a clamp may weigh less than about 0.10 grams, less than about 0.08 grams, or the like, e.g. such that a weight of a clamp may not undesirably interfere with leaflet operation. A clamp may comprise various materials such as, e.g., nitinol, a polymer, a rubber, nylon, stainless steel, nickel titanium, platinum, combinations thereof, or the like.

(50) With reference to FIGS. 7A and 7B, an embodiment of a clamp for clamping a leaflet of a heart valve according to the present disclosure is illustrated, which includes two arms **701**, **702** at a first end **700p** of the clamp. The arms **701**, **702** can transition between a closed configuration in which the arms **701**, **702** are oriented toward each other and/or are aligned substantially in the same plane, and an open configuration in which the arms **701**, **702** are oriented away from each other in different relative planes. In the closed configuration, the second arm **702** extends within the first arm **701**. A spring portion **704** at a second end **700d** of the clamp where each of the arms **701**, **702** connect to each other as part of the same wire filament making up the clamp. The spring portion **704** may be a coiled spring and is configured to bias the arms **702** to the closed configuration. A single wire filament makes up the clamp and extends from the first arm **702** to the spring portions **704** and to the second arm **702** without the wire filament terminating in an end such that there are no sharp portions of the clamp. FIGS. 8A and 8B illustrate first and second arms **801**, **802** having a thinner width profile the arms **701**, **702** of the clamp in FIGS. 7A and 7B. FIGS. 9A and 9B illustrate first and second arms **901**, **902** with the wire filament that makes up the clamp terminating in two ends that make up an end **902p** of the second arm **902**. FIGS. 10A and 10B illustrate a clamp having a first arm **1001** and second arm **1002** at a first end **1000p** of the clamp. The single wire filament that makes up the clamp terminates at an end **1002p** of the second arm **1002**. The clamp includes a spring portion **1004** at a second end **1000d** of the clamp that is configured to bias the arms **1001**, **1002** to the closed configuration.

(51) With reference to FIGS. 11A-11C, an embodiment of a clamp for clamping a leaflet of a heart valve according to the present disclosure is illustrated, which includes a body **1101** with two arms **1102**, **1103** at a first end **1101p** of the body **1101**. The arms **1102**, **1103** have a closed configuration in which the arms **1102**, **1103** are oriented towards each other as illustrated in FIGS. 11A-11C, and an open configuration in which the arms **1102**, **1103** are oriented away from each other. A coiled spring portion **1104** may be at a second end **1101d** of the body **1101**. The spring portion **1104** is configured to bias the arms **1102**, **1103** to the closed configuration. A longitudinal axis **l** may extend through the first end **1101p** and the second end **1101d**. A first cover **1110** may be disposed adjacent to an arm **1102** and may extend along the arm **1102** to the longitudinal axis **l**. A second cover **1112** may be disposed adjacent to the opposing arm **1102** and may extend along the arm **1102** to the longitudinal axis **l**. The covers **1110**, **1112** include protrusions **1106** that extend along the covers **1110**, **1112** and that are configured to embed into tissue. A pin **1114** may extend through the first cover **1110**, the second cover **1112**, the coiled spring portion **1104**, and the longitudinal axis **l**. A first channel **1108** may be disposed on the first cover **1110** extending along the first cover **1100** and has a central axis that may be substantially parallel with an arm **1102**. A second channel **1109** may be disposed on the second cover **1112** extending along the second cover **1112** and has a central axis that may be substantially parallel with the opposing arm **1103**. The channels **1108**, **1109** accommodate the arms **1102**, **1103**, and each channel **1108**, **1109** has a portion without an arm **1102**, **1103** extending away from each of the arms **1102**, **1103** that can accommodate a pin (e.g., a locking pin of a spreader as described below). The covers **1110**, **1112** are rotatable about the pin **1114** to move with the opening and closing of the arms **1102**, **1103**. The covers include a clearance space **1115** near the pin **1114** to allow for this rotation without the covers **1110**, **1112** interfering

with each other.

(52) With reference to FIGS. 12A and 12B, an embodiment of a clamp for clamping a leaflet of a heart valve according to the present disclosure is illustrated, which includes a frame body **1201** with two arms **1202** pivotably connected via a tab **1205**. A first pin **1213** extends through the two arms **1202** and the tab **1205** of the frame body **1201** such that the arms **1202** may rotate about the first pin **1213**. A first cover **1210** may be disposed adjacent to an arm **1202** and may extend along the arm **1202** to a longitudinal axis **l**. A second cover **1212** may be disposed adjacent to the opposing arm **1202** and may extend along the arm **1202** to the longitudinal axis **l**. The covers **1210**, **1212** include protrusions **1206** that extend along the covers **1210**, **1212** and that are configured to embed into tissue. A second pin **1214** may extend through the first cover **1210**, the second cover **1212**, and the longitudinal axis **l**. The arms **1202** have a stable closed configuration, as illustrated in FIG. 12A, in which the arms **1202** are configured for engagement to a tissue, and a stable open configuration, as illustrated in FIG. 12B, in which the arms **1202** are spaced apart from each other. The open configuration may separate the arms **1202** by a given number of degrees apart from each other α , e.g., about 50°, about 60°, about 90°, etc. The clamp transitions between the closed and open configurations by translating the tab **1205** along the longitudinal axis **l** such that the arms **1202** of the frame body **1201** move apart and toward each other along with the covers **1210**, **1212**. As the tab **1205** is moved along the longitudinal axis **l**, the arms **1202** rotate about the first pin **1213** and the covers **1210**, **1212** rotate about the second pin **1214**. The tab **1205** may be manipulated through a tab aperture **1205a** (e.g., via a filament, a wire, a tether, a push member, a catheter, a bowden cable, or the like). The tab **1205** may travel a distance to transition the arms **1202** between the open and closed configurations that may be, e.g., about 0.02 inches (about 0.51 millimeters), or the like. The covers **1210**, **1212** include a clearance space **1215** adjacent the pin **1214** to allow for rotation without the covers **1210**, **1212** interfering with each other.

(53) With reference to FIGS. 13A-13C, an embodiment system for clamping a leaflet of a heart valve according to the present disclosure is illustrated, including a clamp **1300** with arms **1302** at a first end that have a closed configuration and an open configuration, as described throughout this disclosure. The clamp **1300** is in the closed configuration in FIGS. 13A and 13B, and the clamp **1300** is in the open configuration in FIG. 13C. The clamp **1300** has a spring portion **1304** at a second end that biases the arms **1302** to the closed configuration. A spreader **1320** may be releasably coupled to the clamp **1300**. The spreader **1320** may transition the clamp **1300** between the closed configuration and the open configuration. The spreader **1320** includes a base **1322** with a first pin **1324** extending from the base **1322**. A lever **1326** may be rotatably disposed about the first pin **1324**. The base **1322** includes a first channel **1341**, and the lever **1326** includes a second channel **1342**, each channel configured to accept an arm **1302** of the clamp **1300**. The ends of each arm **1302** include an aperture **1308**. Each aperture **1308** has a central axis that extends substantially along each of the arms **1302** and is configured to accept one of two locking pins **1310**. One locking pin **1310** extends through the first channel **1341** and into the aperture **1308** of an arm **1302**, while the other locking pin **1310** extends through the second channel **1342** and into the aperture **1308** of the opposing arm **1302**. The locking pins **1310** fix the arms **1302** to the base **1322** and the lever **1326** such that the spreader **1320** can manipulate the arms **1302** between the open and closed configurations and so that the clamp **1300** cannot be released from the spreader **1320**. Because the arms **1302** are locked within the channels **1341**, **1342**, the arms **1302** may transition between the closed configuration and the open configuration with a movement of the lever **1326**. The lever **1326** may move about the first pin **1324** via translation of a first filament **1330** that may be coupled to a slot **1332** of the lever **1326**. As the lever **1326** rotates about the first pin **1324**, it also moves the arm **1302** that is within the second channel **1342** while the opposing arm **1302** is fixed within the first channel **1341** of the base **1322**. In some embodiments, the pins **1310** may extend through a third and a fourth aperture in each of the base **1322** and the lever **1326** rather than the first and second channels **1341**, **1342**, while still extending into the apertures **1308**. A second filament **1312**

may be coupled at each end to the locking pins **1310** such that the second filament **1312** may be grasped and pulled such that the locking pins **1310** are removed from the apertures **1308** and channels **1341**, **1342**, thereby releasing the arms **1302** from the base **1322** and the lever **1326**. FIG. **13C** illustrates the base **1322** of the spreader **1320** coupled to a distal end of a catheter **1334**. The first filament **1330** extends proximally into the catheter **1334** to be manipulated by a medical professional. Because the base **1320** is coupled to the catheter **1334**, the catheter **1334** may be inserted into a patient to a target location to deliver the clamp **1300**. Also, because the base **1322** is fixed to the catheter **1334**, as the first filament **1330** manipulates the lever **1326** containing an arm **1302**, the lever **1326** works against the bias of the spring portion **1304** and moves the clamp **1300** from the closed configuration to the open configuration by moving the arm **1302** in the lever **1326** away from the fixed arm **1302** in the base **1322**.

(54) Still referring to FIGS. **13A-13C**, an embodiment of a method of clamping a leaflet of a heart valve according to the present disclosure may include inserting the catheter **1334** toward a valve (e.g., through the valve). In some embodiments, the catheter **1334** may include the spreader **1320** disposed on a distal end of the catheter **1334**, which may be reversibly coupled to the clamp **1300** in a closed configuration for navigating through the patient and/or a working channel. Once the catheter **1334** is near the target site of the valve, the clamp **1300** may be transitioned to the open configuration by pulling proximally on the first filament **1330** coupled to the lever **1326** and holding tension on the first filament **1330**. The lever **1326** rotates about the first pin **1324** and moves an arm **1302** within the lever **1326** apart from the opposing arm **1302** in the base **1322**. With the clamp **1300** in the open configuration, the catheter **1334** may move the clamp into position proximate a leaflet (e.g., about a flailing leaflet) of the valve such that an arm **1302** is on either side of the leaflet (e.g., see FIGS. **14A-15** as discussed below). With the clamp **1300** in position about the leaflet, tension may be released on the first filament **1330**, allowing the biased spring portion **1304** of the clamp **1300** to transition the clamp **1300** into the closed configuration about the leaflet. The medical professional may optionally re-open the spreader **1320** and clamp **1300** by again pulling proximally on the first filament **1330** to reposition the clamp **1300** if desired. Repositioning the clamp **1300** may be desirable, e.g., if accidentally released, if a better position is realized after deploying the clamp **1300**, or to configure a tension in an artificial chordae tendineae attached to the clamp **1300**. Once the clamp **1300** is in position, the second filament **1312** attached to the locking pins **1310** may be pulled (e.g., by a grasper, a third filament, or the like) such that the pins **1310** are removed from the apertures **1308** of the arms **1302**. With the pins **1310** removed, the clamp is no longer fixed to the spreader **1320** and the spreader **1320** releases the clamp **1300**. The clamp **1300** may be left delivered on the leaflet, and the catheter **1334** and spreader **1320** may be withdrawn from the patient in an open configuration (e.g., within an outer sheath) or a closed configuration (e.g., by a spring within the spreader **1320** to bias the lever **1326** toward the base **1322**). Additionally, or in the alternative, a clamp may be delivered with a fourth filament (e.g., artificial chordae tendineae) that may extend from the clamp **1300** (e.g., from the spring portion **1304**) to another device (e.g., an anchor) to be used in a medical treatment (e.g., anchoring artificial chordae tendineae to a leaflet and to a papillary muscle).

(55) With reference to FIGS. **14A** and **14B**, the embodiment of a clamp **200** of FIGS. **2A-2E** is illustrated clamping a leaflet **150** of a heart valve according to the present disclosure. The clamp **200** is shown in the closed configuration with its two arms **202** adjacent the leaflet **150**, with one arm **202** on either side of the leaflet **150**. The spring portion **204** at the second end **200d** of the clamp **200** is biasing the two arms **202** toward each other in this closed configuration. In the illustrated embodiment, the clamp **200** has been disposed about the leaflet by leading with the first end **200p** and with the leaflet **150** entering between the two arms **202**. The clamp **200** may have been delivered onto the leaflet **150** using another device, e.g., a spreader, that can articulate the arms **202** between the open configuration for accepting the leaflet **150** and the closed configuration for engaging the leaflet **150**. The clamp **200** may have been unlocked from the spreader for

delivery by, e.g., removing locking pins from the apertures **208**. As the clamp **200** was delivered onto the leaflet **150**, the leaflet **150** may have engaged one or more smooth surfaces **206p** of the protrusions **206**. In the closed configuration, the engaging ends **206d** of most of the protrusions may be at least partially embedded into the leaflet **150**. The embedded protrusions **206** enter less than about 50% of a thickness of a wall of the leaflet **150**. The spring portion **204** is configured to provide a clamping force that compresses the arms **202** and/or protrusions **206** onto the leaflet **150** and may embed the protrusions **206**. In some embodiments, protrusions and/or a spring portion may be configured to embed less than about 25% of a wall of a leaflet, less than 10% of a wall of a leaflet, more than 50% of a wall of a leaflet, or the like, e.g., with reference to FIG. **14C**, the protrusions **206** may not extend into the wall of the leaflet **150** and may instead distort the leaflet between the protrusions **206** such that the clamp is engaged with the leaflet **150** without the protrusions **206** extending into the leaflet. A shape of the clamp, e.g., the spring portion **204** and/or the arms **202** of the clamp, may provide an amount of force that does not pierce the protrusions **206** into and/or through the leaflet **150** leaving room for the leaflet **150** to be distorted within the clamp. In some embodiments, the protrusions may extend a distance from an arm that may be about 0.1 millimeters to about 2.0 millimeters, about 0.5 millimeters to about 1.5 millimeters, about 0.7 millimeters, or the like. In some embodiments, protrusions and/or spring portions may be configured such that a tissue is not punctured by the protrusions so that the clamp may be repositioned from a first delivery site to a second delivery site without significant impact on the first delivery site. In this delivered position, the arms **202** of the clamp **200** are fixedly engaged with the leaflet **150** and the second end **200d** of the clamp **200** may be coupled to a filament, e.g., one or more artificial chordae tendineae, such as described and illustrated with respect to FIG. **15**.

(56) With reference to FIG. **15**, an embodiment of a system for clamping a leaflet **150** of a heart valve according to the present disclosure is illustrated, including four clamps **1500** attached to the leaflet **150**. Each of the clamps **1500** is attached to an end of a filament **1550** that is an artificial chordae tendineae. The filaments **1550** are attached to an anchoring filament **1552** that is further attached to an anchor **1554**. The anchor **1554** is embedded in papillary muscle **152** of the heart **154**. A medical professional may adjust the length and tension of the filaments **1550** and anchoring filaments **1552** such that they may replicate and/or replace chordae tendineae of the heart **154** for function with the leaflet **150** of the valve by varying the length of the filaments **1550**, **1552** between the clamp(s) **1500** and the anchor **1554**. The medical professional may adjust filaments **1550**, **1552** in response to a heart valve regurgitation observation that may be observed via transesophageal echocardiogram and/or fluoroscopy. The filaments **1550**, **1552** are fixed at one end to the leaflet **150** by the clamp(s) **1500** and are fixed at a second end to the papillary muscle **152** by the anchor(s) **1554**. A single anchoring filament **1552** may be coupled to one or more filaments **1550** such that one anchoring filament **1552** and one anchor **1554** may be used for multiple clamps **1500**. In some embodiments, the filaments **1552** and anchoring filaments **1552** may be coupled to one or more clamps **1500** and to each other during delivery of the system into the heart **154**.

(57) FIG. **16** illustrates a perspective view of the clamp in FIGS. **2A-2D** having alternative protrusions **1606a**, **1606b**, **1606c**, and **1606d**, according to an embodiment of the present disclosure. The clamp includes protrusions **1606a**, **1606b**, **1606c**, and **1606d** oriented at various angles that are offset from each other and/or the arms **1601**, **1602**. The first protrusions **1606a** are oriented at a first angle with respect to a longitudinal axis of the first arm **1601**. The second protrusions **1606b** are oriented at a second angle with respect to a longitudinal axis of the second arm **1602** that is larger than the first angle. The third protrusions **1606c** are oriented at a third angle with respect to the longitudinal axis of the second arm **1602** that is larger than the second angle. The fourth protrusions **1606d** are oriented at a fourth angle with respect to the longitudinal axis of the second arm **1602** that is larger than the third angle. Various embodiments may include multiple protrusions at various angles. Some protrusions may engage tissue at alternative angles from other protrusions along the same arm or an opposing arm of a clamp.

(58) All of the devices and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the devices and methods of this disclosure have been described in terms of preferred embodiments, it may be apparent to those of skill in the art that variations can be applied to the devices and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the disclosure. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the disclosure as defined by the appended claims.

Claims

1. A system for clamping a leaflet of a heart valve, said system comprising: a clamp having a first end and a second end; a clamp spreader; and a first filament; wherein: the second end of said clamp is configured to couple to an artificial chordae tendineae; said clamp comprises a first clamp arm and a second clamp arm at the first end of said clamp; said first clamp arm and said second clamp arm have a closed configuration in which said clamp arms are positioned a closed distance with respect to each other, and an open configuration in which said clamp arms are positioned away from each other at an open distance greater than the closed distance; said first clamp arm and said second clamp arm are configured to fixedly engage with a leaflet of the heart valve; said clamp spreader comprises a base and a lever rotatable with respect to said base; said lever is configured to transition said clamp arms between the closed configuration and the open configuration; and said first filament extends from said lever to move said lever with respect to said base to move said clamp arms between the closed configuration and the open configuration.
2. The system of claim 1, wherein: a first channel extends through said base and is configured to accept said first clamp arm; and a second channel extends through said lever and is configured to accept said second clamp arm.
3. The system of claim 2, wherein: a first aperture is defined in said first clamp arm to be aligned with the first channel extending through said base; and a second aperture is defined in said second clamp arm to be aligned with the second channel extending through said lever.
4. The system of claim 3, further comprising: a pin disposed in the aligned aperture and channel of either said first clamp arm and said base, or said second clamp arm and said lever, to couple said clamp and said clamp spreader together; and a release filament coupled to said pin to manipulate said pin to be pulled out of the aligned aperture and channel to release at least one of said first clamp arm from said base, or said second clamp arm from said lever.
5. The system of claim 1, further comprising a second filament coupled to at least one of said first clamp arm or said second clamp arm and manipulable to release said at least one of said first clamp arm or said second clamp arm from said clamp spreader.
6. The system of claim 1, further comprising a catheter having a distal end coupled to said base.
7. The system of claim 6, wherein said catheter is configured for transluminal delivery into a patient.
8. The system of claim 1, further comprising one or more protrusions disposed on one or both of said clamp arms, wherein said one or more protrusions is selected from the group consisting of: barbs, spikes, hooks, and tines.
9. The system of claim 8, wherein said one or more protrusions comprise a plurality of protrusions arranged in a column extending along one of said first and second clamp arms and in a first plane, and a plurality of protrusions arranged in a column extending along the other of said first and second clamp arms and in a second plane different from the first plane.
10. The system of claim 1, wherein said clamp weighs less than 0.08 grams.
11. The clamp of claim 1, wherein said clamp further comprises a spring portion positioned to bias said first and second arms to the closed configuration.

12. A clamp spreader system for transitioning a clamp from an open configuration to a closed configuration to clamp onto a leaflet of a heart valve, said clamp spreader system comprising: a base configured to be coupled with a first clamp arm of a clamp and with a distal end of the catheter; and a lever rotatable with respect to said base, wherein said lever is configured to be coupled with: a second clamp arm of the clamp; and a filament manipulable to manipulate said lever to move the second clamp arm from a closed configuration at a closed distance with respect to the first clamp arm to an open configuration at an opened distance with respect to the first clamp arm which is greater than the closed distance; wherein: a first channel extends through said base and is configured to accept the first clamp arm and to substantially align with an aperture in the first clamp arm and to receive a first pin extended through the aperture in the first clamp arm; and a second channel extends through said lever and is configured to accept the second clamp arm and to substantially align with an aperture in the second clamp arm and to receive a second pin extended through the aperture in the second clamp arm.
13. The clamp spreader of claim 12, wherein said lever has a first end configured for coupling with the filament, and is pivotably coupled to a first end of said base.
14. The clamp spreader of claim 12, further comprising a catheter configured for transluminal delivery into a patient.
15. The clamp spreader of claim 12, further comprising a release filament coupled to said second pin to release said second clamp arm from said lever.
16. A method of clamping a leaflet of a heart valve, said method comprising: inserting a catheter transluminally to the heart valve, the catheter having a clamp spreader system disposed on a distal end of the catheter and reversibly coupled to a clamp, the clamp spreader system comprising: a clamp having a first end and a second end; a clamp spreader; and a first filament; wherein: the second end of the clamp is configured to be coupled to an artificial chordae tendineae; the clamp comprises a first clamp arm and a second clamp arm at the first end of the clamp; the first clamp arm and the second clamp arm have a closed configuration in which the clamp arms are positioned a closed distance with respect to each other, and an open configuration in which the clamp arms are positioned away from each other at an open distance greater than the closed distance; the first clamp arm and the second clamp arm are configured to be fixedly engaged with a leaflet of the heart valve; the clamp spreader comprises a base and a lever rotatable with respect to the base; the lever is configured to transition the clamp arms between the closed configuration and the open configuration; and the first filament extends from the lever to move the lever with respect to the base to move the clamp arms between the closed configuration and the open configuration; positioning the clamp and clamp spreader proximate to the leaflet; and releasing tension on the first filament extending from the lever through the catheter to transition the clamp to a closed configuration about the leaflet.
17. The method of claim 16, said method further comprising: pulling the first filament to move the clamp spreader and the clamp arms into an open configuration; and moving the clamp in the open configuration to a leaflet such that an arm of the clamp is on either side of the leaflet.
18. The method of claim 16, further comprising removing one or more pins from the clamp spreader and clamp such that the clamp spreader releases the clamp.
19. The method of claim 16, further comprising: applying tension to the first filament such that the clamp spreader transitions the clamp from the closed configuration to an open configuration; repositioning the clamp about the leaflet; and releasing tension on the first filament extending through the catheter such that the clamp transitions to the closed configuration about the leaflet.
20. The method of claim 16, further comprising anchoring an artificial chordae tendineae attached to the clamp to a papillary muscle.
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